System Validation Report

Field	Detail
Project	Vehicle Parking System (VPS) v1.1 — Group 6
Generated	07 Nov 2026, 17:21 IST
Environment	Python 3.13.18 · Linux x86_64
Source revision	2f4b768
Prepared by	Himansh Ahuja (QA Lead) · generated by scripts/generate_reports.py

1. Summary

Metric	Value
Test cases in the Test Case Specification	38
Test cases passed	**38 / 38**
Automated tests executed (unit + system + performance)	247
System & performance tests passed	26 / 26
SRS requirements verified	26 fully, 2 partially, of 28
Exit criteria (Test Plan 5.3)	MET: all planned cases executed, no open critical/high defects, security & performance NFRs pass

Validation approach (Test Plan §5): system tests drive complete journeys through the HTTP API (`tests/system`), security validation runs against a live TLS 1.3 server, and performance tests load a separate server process sized to 10,000 bays and 50 lanes (`tests/perf`).

2. Results by test case

Test case	Title	Requirement(s)	Automated tests	Result
TC-ENT-01	ANPR plate capture	VPS-F-001	9/9 passed	**PASS**
TC-ENT-02	QR ticket issue	VPS-F-002, VPS-SR-004	3/3 passed	**PASS**
TC-ENT-03	Entry barrier raise	VPS-F-003	5/5 passed	**PASS**
TC-ENT-04	Ground-loop safety halt	VPS-F-004	5/5 passed	**PASS**
TC-ENT-05	Anti-passback	VPS-F-002	1/1 passed	**PASS**
TC-ENT-06	Invalid / unreadable plate	VPS-F-001	5/5 passed	**PASS**

Test case	Title	Requirement(s)	Automated tests	Result
TC-SLOT-01	Nearest vacant slot by category	VPS-F-005	6/6 passed	**PASS**
TC-SLOT-02	Sensor → Occupied + red LED	VPS-F-006	2/2 passed	**PASS**
TC-SLOT-03	VMS boards match database	VPS-F-007	2/2 passed	**PASS**
TC-SLOT-04	Lot full / concurrent allocation	VPS-F-005	4/4 passed	**PASS**
TC-BILL-01	Tiered fare computation	VPS-F-008	10/10 passed	**PASS**
TC-BILL-02	15-minute grace period	VPS-F-009	5/5 passed	**PASS**
TC-BILL-03	Surge and coupons	VPS-F-010	7/7 passed	**PASS**
TC-PAY-01	Exit scan retrieves invoice	VPS-F-011	1/1 passed	**PASS**
TC-PAY-02	Cashless payment	VPS-F-012	2/2 passed	**PASS**
TC-PAY-03	Exit barrier on PAID only	VPS-F-013	2/2 passed	**PASS**
TC-PAY-04	GST tax receipt	VPS-F-014	4/4 passed	**PASS**
TC-PAY-05	Payment failure & webhook security	VPS-F-012, VPS-F-013	3/3 passed	**PASS**
TC-RES-01	48-hour advance reservation	VPS-F-015	5/5 passed	**PASS**
TC-RES-02	No-show auto-release	VPS-F-016	2/2 passed	**PASS**
TC-RES-03	Reservation check-in at entry	VPS-F-015	4/4 passed	**PASS**
TC-ADM-01	Admin configuration without restart	VPS-F-017	6/6 passed	**PASS**
TC-ADM-02	Report export <= 5 s	VPS-F-018	20/20 passed	**PASS**
TC-Perf-01	Ingress latency p90	VPS-NF-001	1/1 passed	**PASS**
TC-Rel-01	Availability proxy (soak)	VPS-NF-002	1/1 passed	**PASS**
TC-Scale-01	Scale: 10k slots / 50 lanes	VPS-NF-003	1/1 passed	**PASS**
TC-SEC-06	PCI-DSS: no plaintext card data	VPS-NF-004	2/2 passed	**PASS**
TC-Fault-01	Edge offline resilience	VPS-NF-005	8/8 passed	**PASS**
TC-SEC-01	TLS 1.3 enforced	VPS-SR-001	7/7 passed	**PASS**

Test case	Title	Requirement(s)	Automated tests	Result
TC-SEC-02	Password hashing & PII encryption	VPS-SR-002	11/11 passed	**PASS**
TC-SEC-03	RBAC + MFA	VPS-SR-003	29/29 passed	**PASS**
TC-SEC-04	Counterfeit QR rejected	VPS-SR-004	10/10 passed	**PASS**
TC-SEC-05	Immutable audit log	VPS-SR-005	9/9 passed	**PASS**
TC-SYS-01	Walk-in car, UPI payment	VPS-F-001, VPS-F-005, VPS-F-008, VPS-F-012, VPS-F-013	1/1 passed	**PASS**
TC-SYS-02	Reserved EV, short visit	VPS-F-015, VPS-F-009	1/1 passed	**PASS**
TC-SYS-03	Peak-hour card payment with coupon	VPS-F-010, VPS-F-012	1/1 passed	**PASS**
TC-SYS-04	Unreadable plate, cash exit	VPS-F-001, VPS-F-012	1/1 passed	**PASS**
TC-SYS-05	Admin day	VPS-F-017, VPS-F-018	1/1 passed	**PASS**

3. Requirements traceability (updated RTM)

Requirement	Description	Test cases	Status
VPS-F-001	Capture vehicle plate via ANPR within 1.5 s (>= 95 % OCR accuracy)	TC-ENT-01, TC-ENT-06, TC-SYS-01, TC-SYS-04	**VERIFIED**
VPS-F-002	Issue entry ticket with unique QR, signed session ID and timestamp within 2 s	TC-ENT-02, TC-ENT-05	**VERIFIED**
VPS-F-003	Raise entry barrier within 500 ms on valid ticket	TC-ENT-03	**PARTIAL** — RFID-triggered opening not implemented (DEV-07); ticket-triggered opening verified.
VPS-F-004	Prevent barrier descent while vehicle on ground loop	TC-ENT-04	**VERIFIED**
VPS-F-005	Designate nearest vacant slot by vehicle category	TC-SLOT-01, TC-SLOT-04, TC-SYS-01	**VERIFIED**
VPS-F-006	Slot status Occupied + LED red within 2 s of sensor trigger	TC-SLOT-02	**VERIFIED**
VPS-F-007	Refresh VMS vacancy counts within 1 s	TC-SLOT-03	**VERIFIED**
VPS-F-008	Fee by duration, category and tiered rate tables	TC-BILL-01, TC-SYS-01	**VERIFIED**
VPS-F-009	15-minute grace period at zero charge	TC-BILL-02, TC-SYS-02	**VERIFIED**

Requirement	Description	Test cases	Status
VPS-F-010	Peak-hour surge rates and promo coupons	TC-BILL-03, TC-SYS-03	**VERIFIED**
VPS-F-011	Exit QR / plate scan retrieves invoice within 1 s	TC-PAY-01	**VERIFIED**
VPS-F-012	Cashless payment (UPI QR, cards, wallets); webhook PAID within 3 s	TC-PAY-02, TC-PAY-05, TC-SYS-01, TC-SYS-03, TC-SYS-04	**VERIFIED**
VPS-F-013	Raise exit barrier only on verified payment	TC-PAY-03, TC-PAY-05, TC-SYS-01	**VERIFIED**
VPS-F-014	Itemised GST receipt via printer, e-mail or SMS	TC-PAY-04	**VERIFIED**
VPS-F-015	Reserve a slot up to 48 h in advance	TC-RES-01, TC-RES-03, TC-SYS-02	**VERIFIED**
VPS-F-016	Release reserved slot 30 min after start on no-show	TC-RES-02	**VERIFIED**
VPS-F-017	Admin console: tariffs, zones, attendant credentials; no restart	TC-ADM-01, TC-SYS-05	**VERIFIED**
VPS-F-018	Revenue / turnaround / occupancy reports in CSV/PDF within 5 s	TC-ADM-02, TC-SYS-05	**VERIFIED**
VPS-NF-001	Ingress transaction p90 <= 3 s	TC-Perf-01	**VERIFIED**
VPS-NF-002	99.9 % monthly availability	TC-Rel-01	**PARTIAL** — Monthly uptime is measured in production; soak test with zero 5xx is the lab proxy (DEV-10).
VPS-NF-003	10,000 slots, 50 lanes, < 200 ms API latency at 500 req/s	TC-Scale-01	**VERIFIED**
VPS-NF-004	PCI-DSS: no plaintext card data or credentials	TC-SEC-06	**VERIFIED**
VPS-NF-005	72 h autonomous edge queuing; auto-sync <= 5 min	TC-Fault-01	**VERIFIED**
VPS-SR-001	TLS 1.3+ mandatory on all connections	TC-SEC-01	**VERIFIED**
VPS-SR-002	Argon2id/bcrypt passwords; AES-256 PII at rest	TC-SEC-02	**VERIFIED**
VPS-SR-003	RBAC + MFA for administrative accounts	TC-SEC-03	**VERIFIED**
VPS-SR-004	HMAC-SHA256 signed QR tickets; tampered tickets rejected	TC-ENT-02, TC-SEC-04	**VERIFIED**
VPS-SR-005	Immutable write-once audit log retained 3 years	TC-SEC-05	**VERIFIED**

4. Performance & scalability measurements

Measure	Target	Measured	Result
Ingress round trip p90 (200 tx, HTTP) — NF-001	≤ 3000 ms	3.43 ms (p50 2.63, p99 5.46)	PASS

Measure	Target	Measured	Result
Server-side ingress processing p90	—	0.36 ms	info
API latency p95 under load (10,000 bays, 50 lanes, 25 clients) — NF-003	< 200 ms	59.64 ms (p50 23.84, p99 84.32)	PASS
Throughput (closed loop, 1,000 requests) — NF-003	500 req/s	855.8 req/s	PASS
Server errors under load	0	0	PASS
p95 by endpoint	—	ingress 75.33 · sensor 57.77 · VMS 40.91 · nearest-slot 42.97 ms	info
Slowest report export, 5,000-row month — F-018	≤ 5000 ms	83.16 ms	PASS

Measured on the build machine; CI re-runs the suite on every push and publishes `perf-metrics.json`.

5. Security validation (Test Plan 5.4)

Requirement	Validation performed	Result
VPS-SR-001	TLS 1.3 handshake succeeds against live server; TLS 1.2 client refused; plain HTTP not served; HSTS sent	**PASS**
VPS-SR-002	Argon2id hashes; AES-256-GCM ciphertext for plates/contacts/TOTP; DB dump contains no plaintext PII	**PASS**
VPS-SR-003	RBAC matrix across 6 protected endpoints × 4 identities; admin login requires valid TOTP; lockout	**PASS**
VPS-SR-004	Single-character QR tamper and re-signed clone rejected (403) with CRITICAL alert + audit	**PASS**
VPS-SR-005	UPDATE/DELETE refused by DB triggers; out-of-band edit detected by hash-chain verification	**PASS**
VPS-NF-004	No PAN in storage (last-4 only); card numbers Luhn-validated	**PASS**

6. Defects found and fixed during testing

#	Found by	Defect	Fix
1	TC-Scale-01	p95 latency 880 ms / 81 req/s at 10k bays: VMS summary recomputed on every request; BaseHTTPMiddleware overhead; fsync per commit	Versioned summary/VMS cache, pure-ASGI middleware, WAL synchronous=NORMAL → > 500 req/s, p95 ≈ 100–120 ms
2	TC-PAY-04	`/receipts/{id}.pdf` captured by the JSON route (`{id}` matched `abc.pdf`)	PDF route declared first
3	TC-Rel-01	Soak test exits silently failed with 401 after session expiry (test defect)	Soak test uses the cashless path and asserts ≥ 100 exits
4	TC-SLOT-03	Level boards listed categories with no bays (e.g. '2W: 0' on level 3)	Boards show only categories present on the level

No open defects at sign-off.

7. Artefacts

- System/perf HTML report: `docs/reports/html/system-tests.html`
- Deviations: `docs/Deviations.md`
- Test Case Specification: `docs/testing/Test_Case_Specification.md`